IRRIGATION WATER MANAGEMENT · NOTE WM02

Orifices, Weirs and Parshall Flumes — Formulas and Installation

Course topics: IDE5.5 L6, CAE1.3 L11, IDE3.6 L8, IDE-509 T9

Orifices, weirs and Parshall flumes are the most common farm measuring devices: read the water level on a scale fixed to the device and get the discharge from standard formulae, tables or calibration curves. All are accurate enough if properly built, installed and operated; choose by the expected flows and site conditions.

Orifices

- For small streams — border strips, furrows and check basins.
- Sheet-iron or aluminium plates with accurately machined circular or rectangular openings 2.5 to 7.5 cm in diameter; they work under free-flow or submerged-flow conditions.
- A free-flow circular orifice is not sharpened like a weir; fix the scale on the upstream face with zero at the centre of the opening. It works in fairly level channels and needs no stilling pond, unlike weirs.

$$Q = 0.61 \times 10^{-3}\ a\ \sqrt{(2\ g\ h)}$$

Q = discharge (L/s); a = area of the orifice (cm²); g = 981 cm/s²; h = depth of water over the orifice centre (free flow) or the difference between upstream and downstream water levels (submerged), cm.

Submerged orifice: a regular, usually rectangular, sharp-edged opening with level top and bottom edges in a vertical cross wall, with water above its top on both sides; the head is the difference between the upstream and downstream levels, and the same formula applies. It is standard with canal head gates and suits sites with little fall and little floating debris.

> A 5 cm orifice (a = 19.63 cm²) under 10 cm free-flow head: Q = 0.61 × 10^{-3} × 19.63 × √(2 × 981 × 10) = 0.012 × 140.1 ≈ 1.68 L/s.

Weirs

- Measure flow in irrigation channels, the discharge of a well or canal outlet, or flow at any point in a channel.
- A weir is a notch of regular form — a wall or bulkhead of timber, metal or concrete with an opening (the **notch**) of fixed dimensions at the top; rectangular, trapezoidal or triangular; portable or permanent.
- **Pros:** simple, cheap and reliable; one head reading; not easily clogged by moss or trash. **Cons:** need a considerable fall of water surface, so are hard to use on level land; the weir pond may silt up and need clearing.

Term	Meaning
Crest	Bottom edge of the notch
Head (H)	Depth of flow over the crest, measured a set distance upstream
Nappe	Sheet of water falling through the notch
End contractions	Horizontal distances from the crest ends to the channel sides
Free / submerged flow	Free when air reaches all round the nappe; otherwise submerged

General: $Q = C\ L\ H^m$ (C depends on crest and approach; L = crest length; H = head; m depends on the notch shape)

Classification	Types
Notch shape	Rectangular; triangular (V-notch); trapezoidal (Cipolletti)
Crest	Sharp-crested — sharp upstream edge, water touches only a line; used almost exclusively on farms for accuracy. Broad-crested — rounded or broad crest, water touches a surface
Crest length	Contracted — crest shorter than the channel width. Suppressed — crest equal to the channel width

Formulas (Q in L/s, L and H in cm)

Weir	Formula	Notes
Suppressed rectangular (Francis)	Q = 0.0184 L H^(3/2)	Horizontal sharp crest bevelled on the upstream side; for comparatively large flows
Contracted rectangular (Francis)	Q = 0.0184 (L - 0.1 n H) H^(3/2)	n = number of end contractions — usually two, sometimes one
Cipolletti (trapezoidal)	Q = 0.0186 L H^(3/2)	Sides slope 1 horizontal : 4 vertical, so no correction for end contractions; sharp crest and sides bevelled upstream; medium flows
90° V-notch	Q = 0.0138 H^(5/2) (heads ≥ 7.5 cm)	Sides at 45° to the vertical, sharp and bevelled on the downstream side only; small and medium flows, most accurate for small ones; calibrate for heads below 7.5 cm

> A 90° V-notch reading H = 12 cm: Q = 0.0138 × 12^2.5 = 0.0138 × 498.8 ≈ 6.9 L/s. A suppressed rectangular weir with L = 60 cm, H = 10 cm: Q = 0.0184 × 60 × 31.62 ≈ 34.9 L/s.

Standard installation conditions

1. Set the weir at the lower end of a long pool with a smooth current; keep the velocity of approach ≤ 15 cm/s — a weir pond (weir box) is essential, with baffles if needed.
2. Make the weir wall vertical and firmly embedded in the bank, with the weir-box centre line parallel to the flow and the crest level along its whole length.
3. Keep the crest height at least twice the depth of flow over the weir, and the distance from each crest end to the bank about twice the flow depth.
4. Set the crest high enough for a free fall with air under the nappe.

5. Over a rectangular weir keep the depth ≥ 5 cm and ≤ two-thirds of the crest width.
6. Place the gauge about 4H upstream in still water, with zero at crest level (rectangular, trapezoidal) or at the apex (V-notch).
7. Protect the downstream side with stones or brick bats, or a regular stilling basin.

Parshall flume

- Combines the advantages of a weir and a submerged orifice; built of wood, metal or masonry.
- **Sections:** a converging (contracting) inlet with a level floor both ways, a throat whose floor slopes down, and a diverging outlet whose floor slopes up. Its size is the throat width W.
- **Range:** from a very small fraction of a m^3/s to more than 100 m^3/s.
- **Pros:** no upstream pond; approach velocity hardly affects accuracy; does not clog and keeps itself clear of sand and silt; works in shallow channels with gentle grades; tolerates high submergence and a wide range of backwater.
- **Cons:** dearer and harder to build than a weir; only the smallest sizes are portable; cannot be used with turnouts from canals or storage ponds.
- **Heads:** free flow needs only the upstream head H_a; submerged flow needs H_a and the downstream head H_b. Read them in stilling wells with the gauge zero level with the flume crest.

Throat width	Free-flow limit (H_b/H_a)
2.5 – 7.5 cm	0.5
15 – 22.5 cm	0.6
30 – 240 cm	0.7
3 – 15 m	0.8

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Discharge of a 90° V-notch (H in cm, Q in L/s) is

(a) 0.0184 L H^1.5 (b) 0.0186 L H^1.5
(c) 0.0138 H^2.5 (d) 0.61 a√(2gh)

Q2. Side slopes of a Cipolletti weir are

(a) 1:1 (b) 1 H : 2 V
(c) 1 H : 4 V (d) 2 H : 1 V

Q3. Velocity of approach to a measuring weir should not exceed

(a) 5 cm/s (b) 15 cm/s
(c) 50 cm/s (d) 1 m/s

Q4. A Parshall flume with a 30–240 cm throat flows free while H_b/H_a is below

(a) 0.5 (b) 0.6
(c) 0.7 (d) 0.8

Q5. The weir gauge is placed upstream at a distance of about

(a) H (b) 2H
(c) 4H (d) 10H

Q6. A weir whose crest length equals the channel width is

(a) contracted
(b) suppressed
(c) Cipolletti
(d) broad-crested

Answers

Q1 — (c) 0.0138 H^2.5
Q2 — (c) 1 H : 4 V
Q3 — (b) 15 cm/s
Q4 — (c) 0.7
Q5 — (c) 4H
Q6 — (b) suppressed